

One-parameter Λ -cosmology

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Abstract

The standard Λ CDM model specifies the evolution of the Universe through a set of supposedly independent parameters: H_0 , $\Omega_{m,0}$, $\Omega_{\Lambda,0}$, T_0 . We show that a flat dust universe with a cosmological constant is completely determined by the cosmological constant Λ and a dimensionless moment of observation τ_0 . Introducing the dimensionless time $\tau = \frac{3}{2}H_\Lambda t$ (where $H_\Lambda = c/R_\Lambda$, $R_\Lambda = \sqrt{3/\Lambda}$), the entire evolution and the observables at the epoch τ_0 are written through the single function $\sinh(\tau)$: the scale factor, the Hubble parameter, the component densities, the background temperature, and the total causal radius. The vacuum postulate $\bar{\lambda}_\Lambda = \delta R_{\max} = \xi (R_{\max} \ell_P^2)^{1/3} = \sqrt{R_\Lambda \ell_P} = r_s$ at the density-equality epoch $\tau_{m\Lambda}$ uniquely determines $\xi = \sqrt[3]{2/I}$, where $I = \Gamma(\frac{1}{6})\Gamma(\frac{1}{3})/(3\sqrt{\pi}) \approx 2.804$. For a given CMB temperature T_0 (which fixes the moment of observation τ_0) and baryon fraction $\Omega_{b,0}/\Omega_{m,0}$, the model reproduces: $H_0 \approx 67$ km/s/Mpc, the matter fraction $\Omega_{m,0} \approx 0.31$, and the photon-to-baryon ratio $N_\gamma/N_b \approx 1.6 \times 10^9$. In addition, it provides a holographic reformulation of the cosmological constant problem, expressing the observed ρ_Λ through the infrared–ultraviolet scale $\bar{\lambda}_\Lambda = \sqrt{R_\Lambda \ell_P}$ instead of the Planck scale, thereby removing the naive 10^{120} discrepancy. The Universe turns out to be as simple as a non-rotating, uncharged black hole.

Keywords: cosmological constant; vacuum fluctuations; Károlyházy uncertainty; holographic principle; photon-to-baryon ratio

1 Introduction: redundancy of the Λ CDM parameters

The standard Λ CDM model [1] describes the present state of the Universe through a set of observational parameters at the current epoch: the Hubble parameter H_0 , the density fractions $\Omega_{m,0}$, $\Omega_{\Lambda,0}$, and the microwave-background temperature T_0 . These quantities are presented as initial conditions — the data one must measure in order to fix a specific universe out of the family of possible solutions of the Friedmann equations.

In this work we show that these parameters are redundant: a flat Λ CDM universe is fully described by the cosmological constant Λ and the dimensionless moment of time τ_0 at which we observe.

2 Notation and fundamental scales

Let us introduce the key notation, defined by the single parameter Λ and the fundamental constants ($c, \hbar, G, k_B$):

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- t — cosmological time (the age of the Universe);
- $R_\Lambda = \sqrt{3/\Lambda}$ — the de Sitter radius;
- $H_\Lambda = c/R_\Lambda$ — the asymptotic Hubble constant;
- $\rho_\Lambda = \Lambda c^2/(8\pi G)$ — the dark-energy density;
- $T_{\text{dS}} = \hbar H_\Lambda/(2\pi k_B)$ — the de Sitter horizon temperature;
- $\ell_P = \sqrt{\hbar G/c^3}$ — the Planck length;
- $m_P = \sqrt{\hbar c/G}$ — the Planck mass.

The dimensionless time:

$$\tau = \frac{3}{2} H_\Lambda t = \frac{3 c t}{2 R_\Lambda}. \quad (1)$$

3 Evolution through $\sinh(\tau)$

For a flat universe filled with dust and a cosmological constant, the exact solution of the Friedmann equation in the dimensionless time $\tau = \frac{3}{2} H_\Lambda t$ is [2] (see *ibid.* § 5.4.2, Eq. (5.101)). Since after recombination $T \propto a^{-1}$, the entire dynamics of the Universe is thereby expressed through the single function $\sinh(\tau)$ (the coefficients A and B will be substituted in § 9):

$$a(\tau) = A \sinh^{2/3}(\tau), \quad (2)$$

$$H(\tau) = H_\Lambda \cdot \coth(\tau), \quad (3)$$

$$\rho_m(\tau) = \frac{\rho_\Lambda}{\sinh^2(\tau)}, \quad (4)$$

$$T(\tau) = B \sinh^{-2/3}(\tau), \quad (5)$$

$$\Omega_m(\tau) = \text{sech}^2(\tau), \quad (6)$$

$$\Omega_\Lambda(\tau) = \tanh^2(\tau). \quad (7)$$

Equations (3)–(7) contain no free parameters: every observable is a function of the single argument τ , which is determined through Λ and the cosmological time t .

4 Total causal radius and the integral I

Define the *total causal radius* of the Universe as the sum of the particle horizon and the event horizon [3]:

$$R = R_p + R_e = c a \int_0^\infty \frac{dt'}{a(t')}. \quad (8)$$

The physical meaning of this quantity is *the maximal causally accessible scale over the entire span of observation*. The particle horizon R_p sets what a comoving observer is able to see (the past light cone, $\int_0^t dt'/a$), while the event horizon R_e sets what the observer can ever causally influence in the future (the future light cone, $\int_t^\infty dt'/a$). Their sum is the full

comoving region accessible to the observer over the whole history of the Universe, rather than the instantaneous size of the visible part at a fixed epoch. In a dust universe with Λ both integrals converge, so R is finite and well defined. As shown in the previous section (§ 3, Eq. (2)), $a(\tau) = A \sinh^{2/3}(\tau)$. Substituting $dt = (2R_\Lambda/3c) d\tau$,

$$\begin{aligned} R(\tau) &= c a(\tau) \int_0^\infty \frac{dt'}{a(t')} \\ &= c A \sinh^{2/3}(\tau) \int_0^\infty \frac{(2R_\Lambda/3c) d\tau'}{A \sinh^{2/3}(\tau')} \\ &= \frac{2}{3} R_\Lambda \sinh^{2/3}(\tau) \int_0^\infty \frac{d\tau'}{\sinh^{2/3}(\tau')}, \end{aligned} \quad (9)$$

where the coefficient A cancels between the numerator and the denominator of the integral. Hence the dimensionless integral

$$I = \frac{2}{3} \int_0^\infty \frac{d\tau}{\sinh^{2/3}(\tau)}. \quad (10)$$

The substitution $z = e^{-\tau}$ ($\tau = -\ln z$, $d\tau = -dz/z$) together with the identity $\sinh(-\ln z) = (1 - z^2)/(2z)$ leads to a beta integral:

$$\begin{aligned} I &= \frac{2}{3} \int_0^1 \left[\frac{1 - z^2}{2z} \right]^{-2/3} \frac{dz}{z} = \frac{2^{2/3}}{3} \int_0^1 z^{-1/3} (1 - z^2)^{-2/3} dz \\ &= \frac{2^{2/3}}{6} \int_0^1 u^{-2/3} (1 - u)^{-2/3} du \quad (u = z^2) = \frac{2^{2/3}}{6} B\left(\frac{1}{3}, \frac{1}{3}\right) = \frac{2^{2/3}}{6} \frac{\Gamma(\frac{1}{3})^2}{\Gamma(\frac{2}{3})}. \end{aligned} \quad (11)$$

The duplication formula $\Gamma(2z) = 2^{2z-1} \Gamma(z) \Gamma(z + \frac{1}{2}) / \sqrt{\pi}$ at $z = \frac{1}{3}$ and Euler's reflection relation $\Gamma(z) \Gamma(1 - z) = \pi / \sin(\pi z)$ at $z = \frac{5}{6}$ give

$$I = \frac{\Gamma(\frac{1}{6}) \Gamma(\frac{1}{3})}{3\sqrt{\pi}} \approx 2.804. \quad (12)$$

The total causal radius as a function of dimensionless time:

$$R(\tau) = R_\Lambda \cdot I \cdot \sinh^{2/3}(\tau). \quad (13)$$

At the present epoch ($\Omega_{\Lambda,0}/\Omega_{m,0} = \sinh^2 \tau_0 = 2.17$, $\tau_0 = 1.18$), Eq. (13) with the Planck 2018 data gives

$$R_0 = 63.7 \text{ Gly} \quad (14)$$

(for $I = 2.804$). Including the radiation correction $I(\Delta) = 2.766$ yields $R_0 = 62.9$ Gly. For comparison, in standard flat Λ CDM the total causal radius $R_p + R_e$ (particle horizon plus event horizon [3]) at the same Planck parameters gives $R_0^{\text{obs}} = 62.8 \pm 0.6$ Gly.

5 Accounting for the radiation era: correction to the integral I

In the previous sections the total causal radius was computed within the dust model (a universe filled only with dust and a cosmological constant), which is a good approximation for late

epochs. However, at the early stages of the evolution of the Universe radiation dominated (relativistic particles: CMB photons and neutrinos). Let us account for the influence of the radiation era on the causal radius and the integral I .

The Friedmann expansion law including radiation takes the form

$$H^2(a) = H_0^2 (\Omega_{r,0} a^{-4} + \Omega_{m,0} a^{-3} + \Omega_{\Lambda,0}), \quad (15)$$

where $\Omega_{r,0}$ is the radiation fraction at the present epoch ($a_0 = 1$). The total causal radius in comoving coordinates (with $a_0 = 1$) is now

$$R = \frac{c}{H_0} \int_0^\infty \frac{da}{a^2 \sqrt{\Omega_{r,0} a^{-4} + \Omega_{m,0} a^{-3} + \Omega_{\Lambda,0}}} = \frac{c}{H_0} \int_0^\infty \frac{da}{\sqrt{\Omega_{r,0} + \Omega_{m,0} a + \Omega_{\Lambda,0} a^4}}. \quad (16)$$

Performing the standard change of variable $x = a (\Omega_{\Lambda,0}/\Omega_{m,0})^{1/3}$ gives

$$R = R_\Lambda \left(\frac{\Omega_{\Lambda,0}}{\Omega_{m,0}} \right)^{1/3} \cdot I(\Delta), \quad (17)$$

where the corrected integral $I(\Delta)$ has the form

$$I(\Delta) = \int_0^\infty \frac{dx}{\sqrt{\Delta + x + x^4}}, \quad (18)$$

and the dimensionless parameter Δ , characterizing the influence of radiation, is defined by

$$\Delta = \frac{\Omega_{r,0}}{\Omega_{m,0}} \left(\frac{\Omega_{\Lambda,0}}{\Omega_{m,0}} \right)^{1/3}. \quad (19)$$

Using the Planck 2018 data [1] for the present epoch: $\Omega_{m,0} \approx 0.3153$, $\Omega_{\Lambda,0} \approx 0.6847$, the CMB temperature $T_0 \approx 2.7255$ K, and the effective number of neutrino species $N_{\text{eff}} \approx 3.046$, we obtain the radiation density $\Omega_{r,0} \approx 9.21 \times 10^{-5}$. Hence the value of the dimensionless parameter is

$$\Delta \approx 3.78 \times 10^{-4}. \quad (20)$$

Numerical integration gives the value of the corrected integral

$$I(\Delta) \approx 2.7659, \quad (21)$$

which is 1.37% smaller than the base value $I \approx 2.8044$ obtained without radiation.

Thus the influence of the radiation era leads to a small contraction of the total causal horizon of the Universe at early stages by about 1.4%, while the qualitative picture of one-parameter cosmology and of mass constancy is preserved.

6 Total dust mass

Let us show that the total mass of dust matter in the Universe is constant and determined only by Λ .

The dust-matter density ρ_m in a flat universe with a cosmological constant decreases inversely with volume: $\rho_m(\tau) = \rho_\Lambda / \sinh^2(\tau)$. The causal volume of the Universe is defined through the causal radius $R(\tau)$ from Eq. (13):

$$V(\tau) = \frac{4}{3} \pi R^3(\tau) = \frac{4}{3} \pi R_\Lambda^3 I^3 \sinh^2(\tau). \quad (22)$$

Then the total dust-matter mass $M = \rho_m(\tau) V(\tau)$ equals

$$M = \frac{\rho_\Lambda}{\sinh^2(\tau)} \cdot \frac{4}{3} \pi R_\Lambda^3 I^3 \sinh^2(\tau). \quad (23)$$

Substituting $\rho_\Lambda = \frac{3c^2}{8\pi G R_\Lambda^2}$, we obtain

$$M = \frac{c^2 R_\Lambda}{2G} I^3 = M_\Lambda \cdot I^3, \quad (24)$$

where $M_\Lambda = c^2 R_\Lambda / (2G) \approx 1.1 \times 10^{53}$ kg is the Schwarzschild mass for the de Sitter radius R_Λ , and $I^3 \approx 22.05$ is a dimensionless constant. With $R_\Lambda \approx 1.65 \times 10^{26}$ m we obtain $M \approx 2.4 \times 10^{54}$ kg.

All time-dependent factors, as well as the density fractions Ω_m and Ω_Λ , have cancelled. Thus the total dust-matter mass of the Universe is strictly constant in time and uniquely determined by the cosmological constant Λ .

7 The vacuum postulate and the holographic condition

Let us introduce a hypothesis relating the maximal causal scale to the characteristic wavelength of vacuum fluctuations.

Reduced wavelength. When comparing with the Schwarzschild radius r_s it is natural to use the *reduced* wavelength $\bar{\lambda} \equiv \lambda / (2\pi)$: the condition $2\pi r = \lambda$ is written as $r = \bar{\lambda}$. The full wavelength is $\lambda = 2\pi \bar{\lambda}$.

Károlyházy uncertainty. For a spatial region of size R , the quantum uncertainty of length (caused by metric fluctuations) is generically estimated as $\delta R \sim (R \cdot \ell_P^2)^{1/3}$. In our model the dimensionless coefficient ξ fixes this estimate by an exact equality; for the maximal causal horizon $R_{\max}$ we define

$$\delta R_{\max} = \xi \cdot (R_{\max} \cdot \ell_P^2)^{1/3}. \quad (25)$$

The coefficient ξ is determined by the holographic condition (see below); the Schwarzschild radius $r_s = \sqrt{R_\Lambda \ell_P}$ enters the postulate without an additional factor.

The vacuum postulate. The characteristic scale of quantum vacuum fluctuations is set by the reduced wavelength $\bar{\lambda}_\Lambda$. We postulate its exact equality to the Károlyházy uncertainty for $R_{\max}$ and to the Schwarzschild radius:

$$\bar{\lambda}_\Lambda = \delta R_{\max} = \xi \cdot (R_{\max} \cdot \ell_P^2)^{1/3} = \sqrt{R_\Lambda \cdot \ell_P} = r_s, \quad (26)$$

The connection of the scale $\bar{\lambda}_\Lambda = \sqrt{R_\Lambda \ell_P}$ with the infrared horizon size goes back to the work of Cohen, Kaplan, and Nelson [4], and was also discussed by Padmanabhan [5] in the form of the density relation $\rho_\Lambda \approx \sqrt{\rho_{\text{UV}} \rho_{\text{IR}}}$.

Reduced wavelength of vacuum fluctuations.

$$\bar{\lambda}_\Lambda = \delta R_{\max} = \xi \cdot (R_{\max} \cdot \ell_P^2)^{1/3} = \sqrt{R_\Lambda \cdot \ell_P} = r_s \approx 5.2 \times 10^{-5} \text{ m}. \quad (27)$$

The full wavelength is $\lambda_\Lambda = 2\pi\bar{\lambda}_\Lambda \approx 3.3 \times 10^{-4} \text{ m}$. The reduced length $\bar{\lambda}_\Lambda$ is close to the Planck–Einstein scale $\ell_{\text{PE}} = (\hbar G/c^3 \Lambda)^{1/4} \approx 37 \text{ } \mu\text{m}$ (ratio $\bar{\lambda}_\Lambda/\ell_{\text{PE}} \approx 1.4$). As a *hypothetical* illustration we note that in measurements of anomalous quantum noise in YBCO-based Josephson junctions [6] a roll-off of the noise spectral density was observed at $\nu \approx 0.6\text{--}0.8 \text{ THz}$; the corresponding *reduced* wavelength $\bar{\lambda} = c/(2\pi\nu) \approx 60\text{--}80 \text{ } \mu\text{m}$ turns out to be close to $\bar{\lambda}_\Lambda \approx 52 \text{ } \mu\text{m}$ ($\nu \approx 0.92 \text{ THz}$), whereas $\ell_{\text{PE}} \approx 37 \text{ } \mu\text{m}$ would correspond to an overestimated frequency $\approx 1.3 \text{ THz}$. We emphasize that the interpretation of this roll-off as a manifestation of vacuum fluctuations remains controversial and has not received independent confirmation, so we regard this coincidence only as a guide and not as observational proof of the model.

The holographic condition. To compute ξ , let us pass to the special epoch $\tau_{m\Lambda}$ of equality of the matter and dark-energy densities:

$$\Omega_m(\tau_{m\Lambda}) = \Omega_\Lambda(\tau_{m\Lambda}) \implies \sinh(\tau_{m\Lambda}) = 1, \quad \tau_{m\Lambda} = \ln(1 + \sqrt{2}). \quad (28)$$

At this epoch the causal radius equals

$$R_{m\Lambda} = R_\Lambda \cdot I \cdot \sinh^{2/3}(\tau_{m\Lambda}) = R_\Lambda \cdot I, \quad (29)$$

and the background temperature is

$$T_{m\Lambda} = \frac{\hbar c}{2\pi k_B \cdot \xi^3 \cdot I \cdot \sqrt{R_\Lambda \cdot \ell_P}}. \quad (30)$$

Let us impose the requirement that the background temperature $T_{m\Lambda}$ at the epoch $\tau_{m\Lambda}$ exactly equals the Hawking temperature of a black hole with Schwarzschild radius $r_s = \bar{\lambda}_\Lambda = \sqrt{R_\Lambda \cdot \ell_P}$:

$$T_{m\Lambda, \text{Hawking}} = \frac{\hbar c}{4\pi k_B r_s} = \frac{\hbar c}{4\pi k_B \bar{\lambda}_\Lambda}. \quad (31)$$

Equating $T_{m\Lambda} = T_{m\Lambda, \text{Hawking}}$:

$$\frac{\hbar c}{2\pi k_B \cdot \xi^3 \cdot I \cdot \sqrt{R_\Lambda \cdot \ell_P}} = \frac{\hbar c}{4\pi k_B \cdot \sqrt{R_\Lambda \cdot \ell_P}}. \quad (32)$$

Cancelling the common factors, we obtain

$$\xi^3 = \frac{2}{I}. \quad (33)$$

Hence the single dimensionless coefficient of the model is determined analytically:

$$\boxed{\xi = \sqrt[3]{\frac{2}{I}} = \sqrt[3]{\frac{6\sqrt{\pi}}{\Gamma(\frac{1}{6})\Gamma(\frac{1}{3})}} \approx 0.894.} \quad (34)$$

The idea of expressing the CMB temperature through the geometric mean of the Hawking and Planck temperatures is close to the recent approaches of Haug and Tatum [7], who relate the temperature scales of the cosmological and Planck horizons. However, in our work this relation is derived from the Friedmann equations and the holographic condition at the density-equality epoch $\tau_{m\Lambda}$.

8 Numerical estimate and empirical verification

Substituting the data of the Planck 2018 mission [1] shows close agreement of the temperatures:

- Planck+lensing+BAO ($\Lambda \approx 1.106 \times 10^{-52} \text{ m}^{-2}$): $T_{\text{Hawking}} \approx 3.532 \text{ K}$, $T_{m\Lambda} \approx 3.553 \text{ K}$ (deviation **0.59%**).
- Planck-only ($\Lambda \approx 1.089 \times 10^{-52} \text{ m}^{-2}$): $T_{\text{Hawking}} \approx 3.518 \text{ K}$, $T_{m\Lambda} \approx 3.529 \text{ K}$ (deviation **0.31%**).

Sensitivity analysis with respect to the precision of Λ . The measured value of the cosmological constant Λ has a relative experimental uncertainty of about $\delta\Lambda/\Lambda \approx 1.92\%$ (according to Planck 2018: $\Lambda = (1.089 \pm 0.021) \times 10^{-52} \text{ m}^{-2}$). As Λ changes, the ratio of the dark-energy and matter densities at the present epoch $\Omega_{\Lambda,0}/\Omega_{m,0}$ changes. Since the physical matter density is measured very precisely ($\Omega_{m,0}H_0^2 = \text{const}$), the condition of equality of the dust and vacuum densities $\rho_m(z_{m\Lambda}) = \rho_\Lambda$ requires $(1 + z_{m\Lambda})^3 = \rho_\Lambda/\rho_{m,0} \propto \Lambda$. Thus a change in Λ directly shifts the redshift of the equality epoch $z_{m\Lambda} \propto \Lambda^{1/3}$. Correspondingly, the CMB temperature at the equality epoch changes as $T_{m\Lambda} \propto 1 + z_{m\Lambda} \propto \Lambda^{1/3}$.

At the same time the Hawking temperature scales as $T_{m\Lambda,\text{Hawking}} \propto R_\Lambda^{-1/2} \propto \Lambda^{1/4}$. Because both temperatures grow with increasing Λ with close exponents, their ratio $\frac{T_{m\Lambda,\text{Hawking}}}{T_{m\Lambda}} \propto \Lambda^{-1/12}$ has an extremely low sensitivity to variations of Λ .

The sensitivity index (elasticity) of the temperature ratio with respect to a change in Λ is only

$$\frac{d \ln(T_{m\Lambda,\text{Hawking}}/T_{m\Lambda})}{d \ln \Lambda} = -\frac{1}{12} \approx -0.083. \quad (35)$$

Owing to such a small sensitivity, given the experimental uncertainty of the cosmological constant $\delta\Lambda/\Lambda \approx 1.92\%$, the corresponding uncertainty in determining the temperatures is

- for the Hawking temperature: $\delta T_{\text{Hawking}}/T_{\text{Hawking}} = \frac{1}{4} \frac{\delta\Lambda}{\Lambda} \approx 0.48\%$ (absolute uncertainty $\approx 0.017 \text{ K}$);
- for the equality temperature: $\delta T_{m\Lambda}/T_{m\Lambda} = \frac{1}{3} \frac{\delta\Lambda}{\Lambda} \approx 0.64\%$ (absolute uncertainty $\approx 0.022 \text{ K}$);
- for their ratio: $\delta(T_{\text{Hawking}}/T_{m\Lambda})/(T_{\text{Hawking}}/T_{m\Lambda}) = \frac{1}{12} \frac{\delta\Lambda}{\Lambda} \approx 0.16\%$.

Note that the observed discrepancy between the temperatures is only **0.31%** (for Planck-only) and **0.59%** (for the combined Planck+lensing+BAO data). This discrepancy (absolute difference $\Delta T \approx 0.011\text{--}0.021 \text{ K}$) is substantially smaller than the experimental uncertainty in determining the temperatures themselves from Λ ($\approx 0.017\text{--}0.022 \text{ K}$). The fact that the actual discrepancy is smaller than the measurement uncertainty, while the sensitivity of the temperature ratio to variations of Λ is extremely low, convincingly demonstrates the physical consistency, robustness, and observational support of the proposed holographic condition.

Note that within the proposed one-parameter cosmology itself the temperature ratio is identically equal to unity for any universe:

$$\frac{T_{m\Lambda,\text{Hawking}}}{T_{m\Lambda}} = \frac{I \xi^3}{2} \equiv 1. \quad (36)$$

It does not depend on the value of Λ . The dependence $\propto \Lambda^{-1/12}$ arises only under external verification through the standard multi-parameter Λ CDM, where the cosmological constant and the matter density ($\Omega_{m,0}H_0^2$) are treated as independent parameters. The smallness of the exponent ($-1/12$) guarantees a high robustness of this equality against experimental uncertainties.

9 Maximal causal radius and the epoch $\tau_{\max}$

Now all parameters are computed analytically from the cosmological constant Λ and the fundamental constants.

Normalization of the scale factor and the temperature. Let us fix the absolute normalization by the condition of thermal finish: at the most distant epoch $\tau_{\max}$, when $T(\tau_{\max}) = T_{\text{dS}} \approx 2.20 \times 10^{-30}$ K, set $a(\tau_{\max}) = 1$ (causal radius $R = R_{\max}$). In Eqs. (2) and (5) we substitute

$$A = \sinh^{-2/3}(\tau_{\max}), \quad B = \frac{T_{\text{dS}}}{A}. \quad (37)$$

Then the scale factor and the background temperature are written as

$$a(\tau) = \left[\frac{\sinh(\tau)}{\sinh(\tau_{\max})} \right]^{2/3}, \quad (38)$$

$$T(\tau) = \frac{T_{\text{dS}}}{a(\tau)}. \quad (39)$$

The dimensionless time $\tau_{\max}$. The epoch $\tau_{\max}$ is defined as the moment when the CMB temperature drops to the de Sitter temperature: $T(\tau_{\max}) = T_{\text{dS}}$. According to the temperature-evolution formula (39), this corresponds to the scale factor $a(\tau_{\max}) = 1$. Then from the law $T(\tau) \propto 1/a(\tau)$ it follows that

$$T_{\text{dS}} = T_{m\Lambda} \cdot a(\tau_{m\Lambda}), \quad (40)$$

where $\tau_{m\Lambda}$ is the density-equality epoch, at which $\sinh(\tau_{m\Lambda}) = 1$, and hence $a(\tau_{m\Lambda}) = \sinh^{-2/3}(\tau_{m\Lambda})$. Substituting the value of $T_{m\Lambda}$ from the holographic-condition subsection and expressing $\sinh^{2/3}(\tau_{\max})$, we obtain

$$\sinh^{2/3}(\tau_{\max}) = \frac{T_{m\Lambda}}{T_{\text{dS}}} = \frac{1}{\xi^3 I} \sqrt{\frac{R_{\Lambda}}{\ell_P}}. \quad (41)$$

Taking into account the holographic condition $\xi^3 = 2/I$, we finally have

$$\boxed{\sinh^{2/3}(\tau_{\max}) = \frac{1}{2} \sqrt{\frac{R_{\Lambda}}{\ell_P}} \approx 1.6 \times 10^{30}}, \quad (42)$$

which gives $\tau_{\max} \approx 105$. This corresponds to the age of the Universe

$$t_{\max} = \frac{2 R_{\Lambda} \tau_{\max}}{3 c} \approx 1.2 \times 10^{12} \text{ yr} \quad (43)$$

— the thermal finish of the Universe, when the CMB cools down to the de Sitter horizon temperature. This is about 90 times the current age of the Universe.

Maximal causal radius. The corresponding maximal physical causal radius of the Universe at the moment $\tau_{\max}$ equals

$$R_{\max} = R(\tau_{\max}) = R_{\Lambda} \cdot I \cdot \sinh^{2/3}(\tau_{\max}) = \frac{I}{2} \sqrt{\frac{R_{\Lambda}^3}{\ell_P}} \approx 7.5 \times 10^{56} \text{ m}. \quad (44)$$

10 Observable parameters of our epoch

The model gives precise values of the cosmological parameters for our epoch. The cosmological constant Λ fully sets the temperature-evolution curve $T(\tau)$ (through the normalization $a(\tau_{\max}) = 1$ at $T(\tau_{\max}) = T_{\text{dS}}$, see § 9). The moment of observation τ_0 is fixed by the single scale measured today — the CMB temperature $T_0 = 2.7255 \text{ K}$ [1]. From the temperature-evolution law (39), $T(\tau_0) = T_{\text{dS}} [\sinh(\tau_{\max}) / \sinh(\tau_0)]^{2/3} = T_0$, whence

$$\sinh(\tau_0) = \sinh(\tau_{\max}) \left(\frac{T_{\text{dS}}}{T_0} \right)^{3/2}, \quad \tau_0 = 1.18. \quad (45)$$

Substituting into the evolution formulae:

Parameter	Model	Observations
τ_0	1.18	—
t_0	13.8 Gyr	$13.8 \pm 0.02 \text{ Gyr}$
H_0	67 km/s/Mpc	$67.4 \pm 0.5 \text{ km/s/Mpc}$
R_0	63.7 Gly	$62.8 \pm 0.6 \text{ Gly}$
T_0 (input)	2.7255 K	$2.7255 \pm 0.0006 \text{ K}$
$\Omega_{m,0}$	0.31	0.315 ± 0.007
$\Omega_{\Lambda,0}$	0.69	0.685 ± 0.007

The only input of our epoch is the temperature T_0 , which fixes τ_0 ; all the other quantities in the table are predictions. In particular, the matter fraction $\Omega_{m,0} = \text{sech}^2(\tau_0) \approx 0.31$ is not fitted but follows from τ_0 and agrees with the observed value 0.315 ± 0.007 . The curve $T(\tau)$ itself is set by Λ alone (the normalization $a(\tau_{\max}) = 1$ at $T(\tau_{\max}) = T_{\text{dS}}$), so the choice of T_0 merely indicates a point on this curve — the moment at which we observe.

11 Conservation of particle number in the causal volume

The total number of photons and baryons in the causal volume $V(\tau) = \frac{4\pi}{3} R(\tau)^3$ is constant in time. Indeed:

- the dust density falls as $\rho_m \propto \sinh^{-2}(\tau)$;
- the photon density $n_\gamma \propto T^3 \propto a^{-3} \propto \sinh^{-2}(\tau)$;
- the causal volume grows as $V \propto R^3 \propto \sinh^2(\tau)$.

All dependences on τ cancel mutually, and the absolute particle numbers in the causal volume are pure numbers, determined only by R_{Λ}/ℓ_P .

11.1 Number of photons

$$N_\gamma = \frac{\zeta(3)}{24\pi^4} \left(\frac{R_\Lambda}{\ell_P} \right)^{3/2} \approx 3.73 \times 10^{89}. \quad (46)$$

11.2 Number of baryons

$$N_b = \frac{\Omega_b}{\Omega_m} \frac{I^3}{2} \frac{m_P}{m_p} \frac{R_\Lambda}{\ell_P} \approx 2.30 \times 10^{80}, \quad (47)$$

with $\Omega_{b,0}/\Omega_{m,0} \approx 0.156$ (from the Planck 2018 data [1]) and $m_b \approx m_p$, the proton mass.

11.3 Photon-to-baryon ratio

$$\frac{N_\gamma}{N_b} = \frac{\zeta(3)}{12\pi^4} \frac{\Omega_m}{\Omega_b} \frac{m_p}{m_P} \sqrt{\frac{R_\Lambda}{\ell_P}} \approx 1.62 \times 10^9. \quad (48)$$

This ratio is in excellent agreement with the observed value of the baryon asymmetry of the Universe $\eta_{\text{obs}}^{-1} \approx 1.63 \times 10^9$ (corresponding to $\eta \approx 6.12 \times 10^{-10}$ [1]) to better than 1%.

12 Holographic reformulation of the cosmological constant problem

The dark-energy density ρ_Λ can be expressed through the reduced wavelength of the fluctuations $\bar{\lambda}_\Lambda$:

$$\rho_\Lambda = C \frac{\hbar}{c \bar{\lambda}_\Lambda^4}. \quad (49)$$

Let us derive an analytic expression for the coefficient C . The dark-energy density is related to the cosmological constant by $\rho_\Lambda = \frac{\Lambda c^2}{8\pi G}$. Expressing Λ through the de Sitter radius $R_\Lambda = \sqrt{3/\Lambda}$, we have

$$\rho_\Lambda = \frac{3c^2}{8\pi G R_\Lambda^2}. \quad (50)$$

Using $\bar{\lambda}_\Lambda = \sqrt{R_\Lambda \ell_P}$ and $\ell_P^2 = \frac{\hbar G}{c^3}$, we write

$$\bar{\lambda}_\Lambda^4 = R_\Lambda^2 \ell_P^2 = R_\Lambda^2 \frac{\hbar G}{c^3}. \quad (51)$$

Expressing from this $GR_\Lambda^2 = \bar{\lambda}_\Lambda^4 c^3 / \hbar$ and substituting into the formula for ρ_Λ , we obtain

$$\rho_\Lambda = \frac{3}{8\pi} \frac{\hbar}{c \bar{\lambda}_\Lambda^4}. \quad (52)$$

Thus the exact analytic coefficient is

$$C = \frac{3}{8\pi} \approx 0.1194. \quad (53)$$

Consequently, the vacuum density is expressed through the reduced wavelength $\bar{\lambda}_\Lambda$ as follows:

$$\rho_\Lambda = \frac{3}{8\pi} \frac{\hbar}{c \bar{\lambda}_\Lambda^4}. \quad (54)$$

Passing to the vacuum energy density $E_\Lambda = \rho_\Lambda c^2$, we obtain the exact quantum relation

$$E_\Lambda = \frac{3}{8\pi} \frac{\hbar c}{\bar{\lambda}_\Lambda^4}. \quad (55)$$

Numerically, with $\bar{\lambda}_\Lambda \approx 52 \mu\text{m}$ (from the Planck data), these formulae reproduce the observed ρ_Λ .

This relation *reformulates* the cosmological constant problem in a holographic way: the observed ρ_Λ is naturally expressed through the infrared–ultraviolet scale $\bar{\lambda}_\Lambda = \sqrt{R_\Lambda \ell_P}$, and the naive 10^{120} discrepancy that arises when cutting off at the Planck scale ℓ_P does not appear. We emphasize that Λ remains the single input parameter (formula (54) is an exact rewriting of $\rho_\Lambda = 3c^2/(8\pi G R_\Lambda^2)$); however, the cutoff scale $\bar{\lambda}_\Lambda$ is *not arbitrary*. It is uniquely set by the Károlyházy uncertainty: $\bar{\lambda}_\Lambda = \delta R_{\text{max}} = \xi (R_{\text{max}} \ell_P^2)^{1/3}$ is the minimal geometrically resolvable length for the maximal causal horizon R_{max} . The vacuum is cut off not at the Planck length ℓ_P , but at this minimal scale of geometry admissible for a region of size R_{max} — which agrees with the approach of Cohen–Kaplan–Nelson [4] and Padmanabhan [5], and is precisely why it reproduces the observed ρ_Λ .

Note that in modern theoretical physics there is no single commonly accepted value for the dimensionless coefficient relating the vacuum energy density to the characteristic quantum-cutoff wavelength λ . Usually in quantum-field-theory computations this relation has the form $E_{\text{vac}} \sim \hbar c/\lambda^4$, where the coefficient C depends on the chosen regularization scheme. For instance, for a sharp momentum cutoff of the free electromagnetic field at the scale $k_{\text{max}} = \pi/\lambda$ (corresponding to the Nyquist limit for a lattice step $\lambda/2$) the coefficient is $C = \pi^2/4 \approx 2.47$, whereas at $k_{\text{max}} = \pi/(2\lambda)$ it equals $C = \pi^2/64 \approx 0.154$, and for a scalar field at $k_{\text{max}} = 1/\lambda$ it equals $C = 1/(8\pi^2) \approx 0.013$. In geometrically bounded systems, such as a metallic cube of side L in the Casimir effect, the vacuum energy density is determined by the exact coefficient $C \approx 0.092$. Our analytically derived coefficient $C = 3/(8\pi) \approx 0.119$ naturally falls within the physical range of QFT and the Casimir effect ($10^{-2} \div 10^{-1}$), while not being a fitting parameter. In contrast to the dimensional scale of Beck and de Matos [8], where the coefficient is taken equal to unity by definition (which gives the scale $37 \mu\text{m}$ and a noise cutoff frequency $\approx 1.3 \text{ THz}$), our value $\bar{\lambda}_\Lambda \approx 52 \mu\text{m}$ corresponds to a lower frequency $\nu \approx 0.92 \text{ THz}$. If the roll-off of anomalous quantum noise in YBCO junctions [6] is indeed related to vacuum fluctuations — which remains an unverified and debatable hypothesis — then our scale agrees with it better; however, we do not rely on this experiment as confirmation of the model.

13 Conclusion

Under the vacuum postulate $\bar{\lambda}_\Lambda = \delta R_{\text{max}} = \xi (R_{\text{max}} \ell_P^2)^{1/3} = \sqrt{R_\Lambda \ell_P} = r_s$ and the holographic condition at the density-equality epoch ($\xi = \sqrt[3]{2/I}$), the flat dust ΛCDM universe turns out to be strictly one-parameter in its dynamics. Below we discuss the consequences for the observable quantities and for the classical cosmological problems.

13.1 One-parameter character

The single fundamental parameter Λ sets the dust mass, $H(\tau)$, the fractions $\Omega_m(\tau)$ and $\Omega_\Lambda(\tau)$, the causal radius $R(\tau)$, the CMB temperature $T(\tau)$, the numbers N_γ and N_b , as well as $\tau_{\max} \approx 105$. The four customary “inputs” of Λ CDM — H_0 , $\Omega_{m,0}$, $\Omega_{\Lambda,0}$, T_0 — are not independent initial conditions but the values of the same functions at the point $\tau = \tau_0$. Instead of four free parameters, two quantities remain: Λ and the moment of observation τ_0 .

13.2 The cosmological constant problem

The standard 10^{120} discrepancy between the naive vacuum estimate (with the Planck scale of fluctuations) and the observed density ρ_Λ is *reformulated* rather than resolved from first principles in our construction: if the characteristic scale of vacuum fluctuations is taken to be not ℓ_P but $\bar{\lambda}_\Lambda = \sqrt{R_\Lambda \ell_P} = r_s \approx 52 \mu\text{m}$, then the observed ρ_Λ is reproduced automatically. This is a holographic (IR–UV) reformulation in the spirit of [4, 5], and the cutoff scale is not chosen arbitrarily: it is set by the Károlyházy uncertainty $\bar{\lambda}_\Lambda = \delta R_{\max}$ — the minimal resolvable geometry of the maximal causal horizon. Therefore the cutoff of the vacuum at this scale, rather than at the Planck scale, is a consequence of the quantum uncertainty of spatial length, not a fit.

13.3 The photon-to-baryon ratio

From the constancy of the total numbers of photons and baryons in the causal volume (formulae (46)–(47)) follows the photon-to-baryon ratio (48). It is derived analytically from Λ through $R_\Lambda = \sqrt{3/\Lambda}$ and is not an independent parameter of the model: for fixed Λ and the measured baryon fraction $\Omega_{b,0}/\Omega_{m,0}$ (Planck 2018) we obtain $N_\gamma/N_b \approx 1.62 \times 10^9$, in agreement with the observed baryon asymmetry $\eta_{\text{obs}}^{-1} \approx 1.63 \times 10^9$ ($\eta \approx 6.1 \times 10^{-10}$) to better than 1%.

13.4 The fine-tuning problem

The problem is absent: for fixed Λ there is no arbitrary choice of H_0 , $\Omega_{m,0}$, or $\Omega_{\Lambda,0}$ — they are uniquely determined by τ_0 (and conversely, from the measured $\Omega_{m,0}$ one finds τ_0). Likewise T_0 , N_γ , and N_b are fixed.

13.5 Analogy with a black hole

In its dynamics the Universe is as simple as a non-rotating Schwarzschild black hole, specified by a single mass M : here this role is played by Λ . The epoch of observation τ_0 plays the role of “where on the orbit we look” — not new physics, but a choice of moment on an already-fixed solution.

$$\boxed{\text{Universe} \equiv \Lambda.} \tag{56}$$

References

- [1] Planck Collaboration, N. Aghanim, Y. Akrami, M. Ashdown, et al. Planck 2018 results. VI. cosmological parameters. *Astronomy & Astrophysics*, 641:A6, 2020.

- [2] Barbara Ryden. *Introduction to Cosmology*. Cambridge University Press, 2nd edition, 2017.
- [3] Tamara M. Davis and Charles H. Lineweaver. Expanding confusion: Common misconceptions of cosmological horizons and the superluminal expansion of the universe. *Publications of the Astronomical Society of Australia*, 21(1):97–109, 2004.
- [4] Andrew G. Cohen, David B. Kaplan, and Ann E. Nelson. Effective field theory, black holes, and the cosmological constant. *Physical Review Letters*, 82(25):4971–4974, 1999.
- [5] T. Padmanabhan. Cosmological constant: The weight of the vacuum. *Classical and Quantum Gravity*, 22(16):L107–L112, 2005.
- [6] C. Hoffmann et al. Anomalous quantum noise in $\text{YBa}_2\text{Cu}_3\text{O}_{7-\delta}$ josephson junctions. *Physical Review B*, 70(18):180503(R), 2004.
- [7] Espen Gaarder Haug and J. M. Tatum. The CMB temperature as a geometric mean of the Hawking and Planck temperatures. *Preprint at ResearchGate*, 2023.
- [8] C. Beck and C. J. de Matos. The Planck-Einstein scale: A natural scale for dark energy? *arXiv preprint arXiv:0709.2373*, 2007.